



Effect of Bisoprolol Versus Other Beta-Blockers on Glycemic Control and Metabolic Parameters in Type 2 Diabetes: A Retrospective Cohort Study

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Abstract

Purpose β -blockers (BB) are a heterogeneous class of medications guideline-recommended for heart failure (HF) and diabetes mellitus (DM). However, physicians may be reluctant to prescribe BBs due to concerns regarding negative metabolic effects. The primary objective was to assess the short-term change in glucose control in PwT2DM (patients with T2DM) who initiated bisoprolol versus the reference cohorts. Secondary objectives were to assess change in lipid control, and proportion of patients with hypoglycemic events during follow-up in PwT2DM who initiated bisoprolol.

Methods We performed a retrospective cohort study of T2DM patients on bisoprolol compared to no treatment, carvedilol, other selective BBs, or nonselective BBs. Anonymized data was collected from the Maccabi Health Services (MHS) database between 2003 and 2023.

Results A total of 12,981 records were analyzed. The reduction in HbA1C (difference of -0.05 [-0.09 , -0.01]), fasting plasma glucose (FPG) (difference of -1.72 [-3.35 , -0.1]), and cholesterol measures (total, HDL, LDL differences of -9.12 [-10.94 , -7.31], -0.94 [-1.24 , -0.65], and -7.89 [-9.53 , -6.26], respectively) was significantly greater in the bisoprolol cohort compared to reference cohort 1 (no treatment). In cohort 2, the reduction in HbA1C was significantly greater with carvedilol users compared to bisoprolol users (difference of 0.34 [0.04 , 0.63]). The reduction in total cholesterol and LDL was significantly greater in the bisoprolol cohort compared to reference cohort 4 (nonselective BB users, differences of -5.86 [-9.46 , -2.25] and -7.08 [-10.25 , -3.92], respectively).

Conclusion Bisoprolol had a neutral effect on the glycemic profile with a statistically beneficial impact on the lipid profile, comparable to carvedilol.

Keywords β -blockers · Heart failure · Diabetes · HbA1C · Lipid profile

Introduction

β -blockers (BB) are a class of medications commonly used to treat congestive heart failure (CHF), atrial fibrillation (AF), coronary artery disease (CAD), and hypertension

(HTN) via a decrease of heart contractions and heart rate [1]. BBs do not represent a homogeneous class and show several pharmacological differences, among which β 1-selectivity and additional direct vasodilating property are of special interest [2]. Some BBs such as bisoprolol, metoprolol, or

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nebivolol are selective for β_1 adrenoceptors found mainly in the heart as opposed to β_2 receptors that contribute to dilation of vascular and airway smooth muscle. Bisoprolol has the highest β_1 receptor selectivity (14-fold more selective for β_1 over β_2 receptors) [3]. Others, such as carvedilol, are nonselective for β_1 , β_2 , and have an additional effect on α receptors [1].

Guidelines of the European Society of Cardiology (ESC) support the use of β -blockers in patients with heart failure (HF) [4]. The most recent joint guidelines of the ESC/European Association for the Study of Diabetes (ESC/EASD) recommend β -blockers (class IA recommendation) as first-line therapy for heart failure with reduced ejection fraction (HFrEF) to reduce the risk of hospitalizations for HFrEF and all-cause death; similarly, β -blockers are recommended in patients with DM after acute myocardial infarction (AMI) with a left-ventricular ejection fraction < 40% to prevent sudden cardiac death (class IA recommendation) [5]. The 2017 American Heart Association (AHA) HTN guidelines, as well as the European Society of Hypertension (ESH) 2023 HTN guidelines, recommend β -blockers as combination therapy in patients not controlled on monotherapy for appropriate patient groups, including those with ischemic heart disease (IHD) and HTN [2, 6]. Despite these recommendations, and their proven clinical benefits [7], some physicians are still reluctant to prescribe BB in patients with diabetes due to perceived negative metabolic effects, which were demonstrated with older agents [8].

In light of the conflicting evidence of BB and their metabolic effects, the primary objective of this study was to reexamine the specific effect of bisoprolol on glycemic control and lipid profile in patients with T2DM. To allow clinical framework of reference, we compared T2DM patients treated with bisoprolol to those with no treatment, those treated with carvedilol as a specific nonselective BB with vasodilating properties, patients treated with any nonselective BBs, and individuals treated with any other selective BBs. Secondary objectives were to assess change in lipid control and proportion of patients with at least one hypoglycemic event during follow-up in PwT2DM who initiated bisoprolol. An additional secondary objective was to determine the effect of bisoprolol on glycemic status in patients with specific demographic or risk factor variables. Analysis was done both on the entire follow-up period, and separately on the first and last 6 months.

Methods

Data Source

This study was submitted to the Maccabi Institutional Ethical Review Board (IRB) and approved (approval number

0001–23-MHS). All data were de-identified prior to computational analysis, did not leave Maccabi's infrastructure, and was not intended to impact or monitor ongoing patient care (non-interventional and non-observational). Patient consent was waived owing to the de-identified and retrospective nature of the study. According to the Israel Ministry of Health regulations, individual-level data cannot be shared openly. Specific requests for remote access to de-identified community-level data or code used should be referred to Maccabi Healthcare Services Research and Innovation Center.

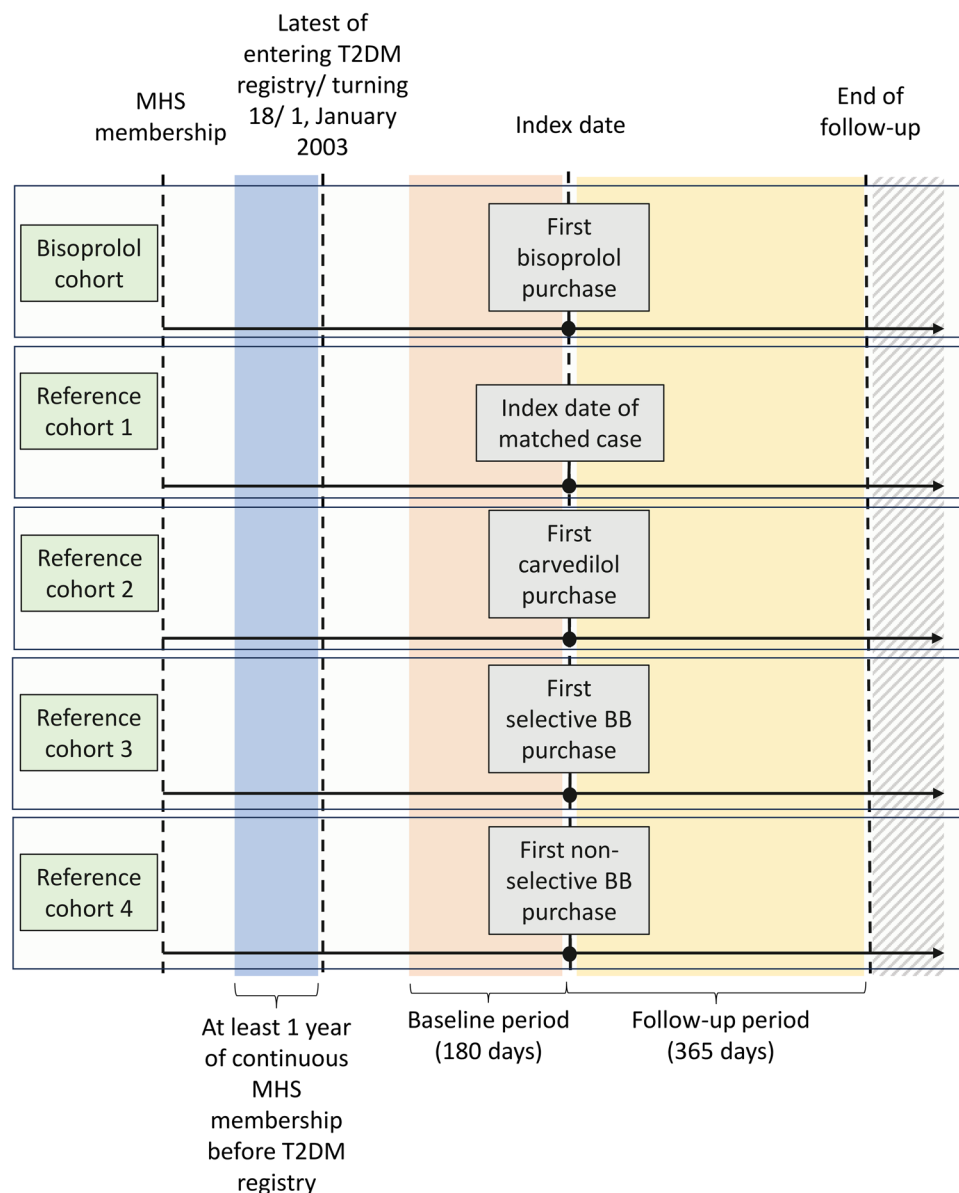
This retrospective cohort study was conducted using de-identified electronic health records (EHRs) of member of Maccabi Health Services (MHS). MHS is the second-largest state-mandated, not-for-profit, health fund in Israel, serving 26.7% of the nation's population, as a representative sample of the Israeli population. Membership in one of the existing four national health funds is mandatory, though residents can freely select the fund of their choice. By law, health funds are not permitted to refuse membership to any individual. MHS has managed a centralized HER database for more than three decades, with a membership disengagement rate of less than 1%, enabling extensive longitudinal medical monitoring. This centralized dataset comprises extensive demographic information, clinical measurements, outpatient and inpatient diagnoses and procedures, prescribed and purchased medications, imaging, and detailed laboratory data from a single central lab.

Study Population

The key inclusion criteria included patients who were 18 years or older and were MHS members with non-insulin-dependent T2DM between 1 January 2003 and 31 December 2023 (inclusive), with at least 1 year of continuous membership in MHS prior to T2DM diagnosis (Fig. 1). Eligible patients must have maintained a stable antidiabetic regimen for at least 90 days prior to the index date or received no antidiabetic treatment during that period. Additionally, the inclusion criteria required at least one HbA1c measurement during the baseline period (180 days prior to the index date) and at least one HbA1c measurement during the follow-up period (between 90- and 365-days post index date). The exclusion criteria included patients with first bisoprolol or other BB purchase prior to T2DM diagnosis, female patients during pregnancy during the follow-up period, patients treated with insulin, GLP-1 RA, or undergoing bariatric surgery during follow-up, and patients with any corticosteroid purchase during the baseline period or follow-up, due to its glycemic effect.

The study included four axes of comparison, each consisting of a “bisoprolol cohort” and a “reference cohort.” Reference cohort 1 included patients without any documentation

Fig. 1 Patient timeline before and during the study. All patients in all four cohorts were registered with T2DM and older than 18 between 1 January 2003 and 31 December 2023, with at least 1 year of continuous NHS membership before entering the diabetes registry. Index date was defined differently for each cohort: First purchase of bisoprolol/carvedilol/selective BB/nonselective BB for the bisoprolol cohort and reference cohorts 2, 3, and 4, respectively, and the index date of the matched bisoprolol user for patients in reference cohort 1



of BB purchase during the follow-up period. Reference cohort 2 included patients with a documented first purchase of carvedilol, after T2DM diagnosis and without any documentation of bisoprolol or any other BB purchase. Reference cohort 3 included patients with a documented first purchase of a β_1 selective BB after T2DM diagnosis, but not of bisoprolol or any other BB purchase. Reference cohort 4 included patients with a documented first purchase of nonselective BB, after T2DM diagnosis and without any documentation of bisoprolol or any β_1 selective BB purchase. Classification of BB is shown in Supplementary Table S1. For all reference cohorts nonexposed patients were matched by birth year, sex, and year of T2DM onset at a 1:1 ratio to bisoprolol exposed patients. Of note, eligible bisoprolol patients might be repeated between different

axes of comparison, but not within each of the four axes of comparison.

Study Variables

De-identified EMR data were retrieved from the centralized MHS database for the study period of January 2003 to January 2023. Patient-level data included demographics, namely biological sex, age, social sector, and socioeconomic status (SES). The SES, assigned by Israel's Central Bureau of Statistics (CBS) [9], is measured on a scale from 1 (lowest) to 10, and the index is calculated on the basis of several parameters, including household income, educational qualifications, household crowding, and car ownership. Data were also collected on chronic diseases from MHS' automated

registries, including cardiovascular diseases [10], diabetes [11], inflammatory bowel diseases, and cancer from the National Cancer Registry (NCR) [12].

The variables for short-term change in glucose control included HbA1c and fasting plasma glucose (FPG). The variables to assess possible change in lipid control included total-cholesterol, LDL-cholesterol, HDL-cholesterol, and triglycerides. If a patient had more than one measurement, the most recent measurements during the baseline and follow-up periods were selected.

Additionally, for the analysis on proportion of patients with hypoglycemic events, hypoglycemic events were defined using the following ICD-9 codes: 251.0 (hypoglycemic coma), 251.1 (other specified hypoglycemia), 251.2 (hypoglycemia, unspecified), and 962.3 (poisoning by insulins and antidiabetic agents). The event was captured if it occurred during follow-up.

Statistical and Analytical Methods

In each axis of comparison, exact matching of bisoprolol users to the reference population was performed by sex, year of birth, and T2DM onset at a 1:1 ratio. To create comparable populations, the inverse probability of treatment weighting (IPTW) method was used. Each patient's probability of being exposed to bisoprolol was calculated using logistic regression, and stabilizing weights were applied in modeling changes in HbA1c and other measurements. Extreme IPTW values were excluded as outliers. Baseline characteristics differences were assessed using standardized mean difference (SMD), with an SMD < 0.1 indicating negligible differences. Baseline and follow-up values were collected and the unadjusted means and standard deviations of the individual differences between these values over each cohort are presented.

Differences in changes between cohorts were compared using linear regression, adjusted for baseline characteristics with SMD > 0.1 after IPTW.

To further investigate the potential impact of bisoprolol on glycemic control, we employed an analytical approach to ensure that no specific subpopulations are at risk. We utilized a linear model to evaluate the interaction between bisoprolol usage and various covariates, specifically focusing on interactions with age, obesity, and baseline HbA1c levels. The absolute change in HbA1c was considered as the primary outcome variable.

In the covariate selection phase, we applied LASSO (least absolute shrinkage and selection operator) regularization to identify and remove nonsignificant covariates from the model. LASSO is a regression analysis method that performs both variable selection and regularization to enhance prediction accuracy and interpretability. We set the regularization parameter, alpha, to 0.05, controlling the strength

of the penalty applied to the coefficients. Higher values of alpha lead to more coefficients being shrunk to zero. During the LASSO procedure, each covariate's coefficient is penalized, and those not contributing significantly to the model's predictive power are reduced to zero, which helps in refining the model by focusing on variables that contribute meaningfully to the outcome.

After covariate selection, we fitted the linear model without regularization to extract the coefficients, confidence intervals (CIs), and *p* values for the remaining variables. This step involved estimating the relationship between the outcome variable (change in HbA1c) and the covariates that were not eliminated by the LASSO method. We specifically assessed the interactions between bisoprolol usage and each covariate to determine if the effect of bisoprolol on glycemic control differs among these subpopulations. We examined the coefficients and significance of variables and interaction terms to identify any significant effects of bisoprolol on glycemic control in conjunction with age, obesity, or baseline HbA1c levels. Significant interaction terms would suggest that the effect of bisoprolol on glycemic control is modified by these covariates, indicating potential at-risk subpopulations.

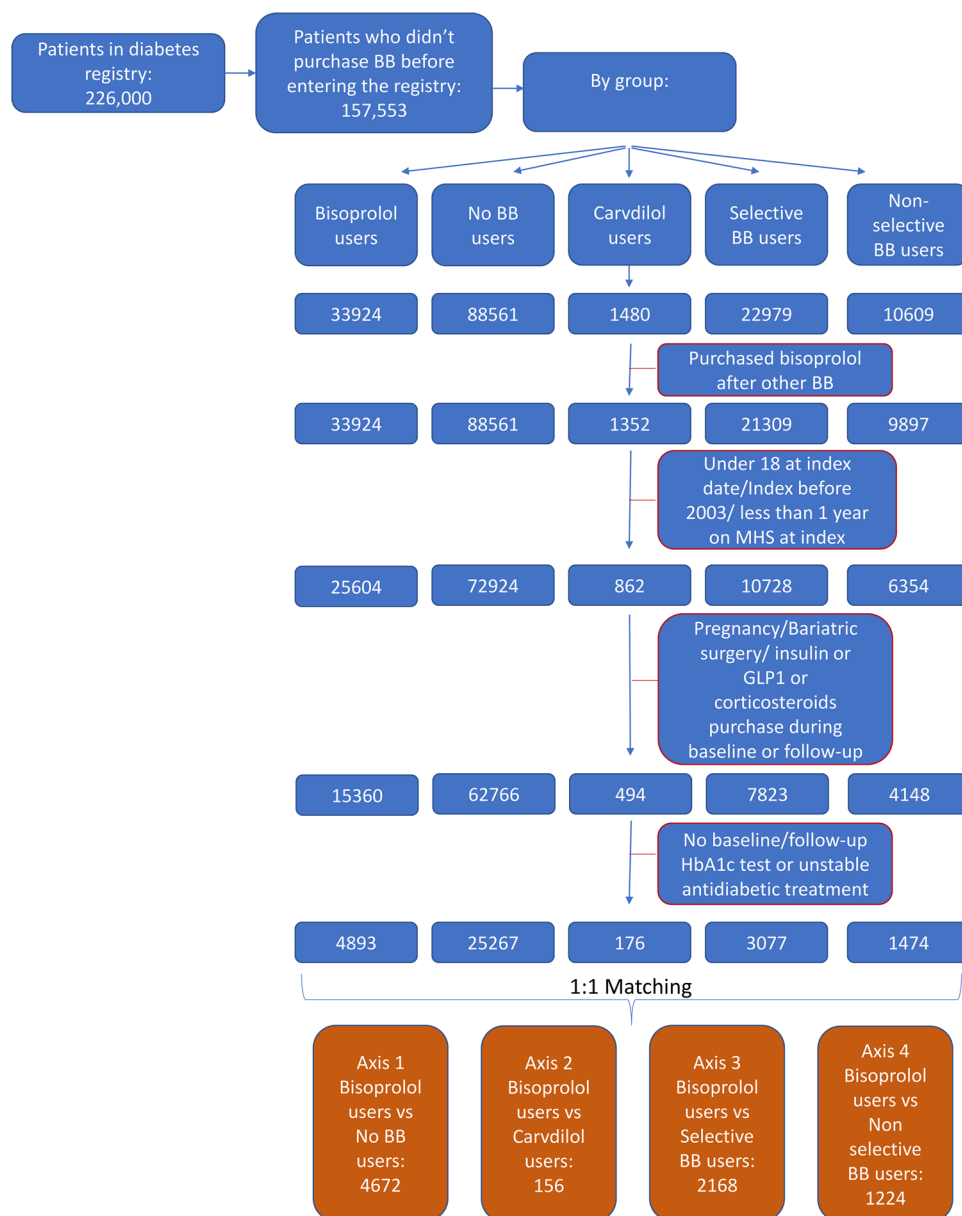
Multiple hypothesis testing was not performed because the focus was on clinically significant differences. Python version 3.10.9 was used for analysis.

Results

Study Patients

A total of 226,000 unique MHS members were found to be diagnosed with T2DM through the validated diabetes registry (Fig. 2). Out of these, 157,553 individuals did not purchase BB before entering the registry. After applying the inclusion and exclusion criteria, the study population included 25,267 individuals who did not purchase any BB, 4893 people who purchased bisoprolol, 176 carvedilol users, 3077 selective BB users, and 1474 individuals who purchased nonselective BB. We then matched the bisoprolol users in each axis to the individuals in the appropriate reference cohort for gender, birth year, and year of T2DM diagnosis, a total of 12,981 patients' records were analyzed. Axis 1 comprised 4672 patients each in the bisoprolol cohort and reference (no BB purchase) cohort 1 (median age of 64 (57–71)). The median duration of T2DM at index date was 6 (2–9) years. Axis 2 comprised 156 patients each in the bisoprolol cohort (median age of 68 (59.0–77.2) and reference cohort 2 (carvedilol users, median age of 67 (58–76)). The median duration of T2DM at index date was 6 (4–9) in reference cohort 2 and 8 (5–12.2) in the bisoprolol cohort. Axis 3 comprised

Fig. 2 Cohort selection flowchart for the comparative analysis of β -blocker use among patients with type 2 diabetes in the Maccabi Healthcare Services (MHS) diabetes registry. Starting from 226,000 patients in the diabetes registry, 157,553 patients without prior β -blocker (BB) use were included. These patients were categorized into groups based on subsequent BB use: bisoprolol users ($n = 33,924$), no BB users ($n = 88,561$), carvedilol users ($n = 1,480$), selective BB users ($n = 22,979$), and nonselective BB users ($n = 10,609$). Stepwise exclusion criteria were applied, including age < 18 at index date, index date before 2003, < 1 year in MHS, pregnancy, bariatric surgery, insulin/GLP-1/corticosteroid use, and missing HbA1c data or unstable antidiabetic treatment. Final matched cohorts were generated for comparisons along four axes: bisoprolol vs. no BB users ($n = 4,672$), carvedilol users ($n = 156$), selective BB users ($n = 2,168$), and nonselective BB users ($n = 1,224$).



2168 patients each in the bisoprolol cohort (median age of 66 (59–73)) and reference cohort 3 (selective BB users, median age of 62 (56–69.25)). The median duration of T2DM at index date was 5 (2–7) in reference cohort 3 and 8 (5–12) in the bisoprolol cohort. Axis 4 comprised 1224 patients each in the bisoprolol cohort (median age of 67 (59.75–73)) and reference cohort 4 (other nonselective BB, median age of 66 (58–72)). The median duration of T2DM at index date was 6 (3–9) in reference cohort 4 and 7 (4–11) in the bisoprolol cohort. Demographic and clinical data at baseline can be found in Supplementary Tables S2 a–d.

Absolute Changes

The changes from baseline to follow-up in HbA1C, FPG, cholesterol, LDL-cholesterol, HDL-cholesterol, and triglycerides in all cohorts were examined (Tables 1, 2, 3, and 4). HbA1C dropped in all axes of comparison for both cohorts, but the reduction was significantly greater in reference cohort 2 (carvedilol users, baseline 7.266, follow-up 6.876, bisoprolol users, baseline 7.04, follow-up 6.987, $p = 0.024$). When compared to reference cohort 1, where there was no documented use of BB, bisoprolol users had a significantly greater drop in HbA1C (no BB use, baseline 6.857, follow-up 6.819,

Table 1 Absolute change in measurements, bisoprolol vs. reference cohort 1 (no documentation of BB purchase)

Measurement	Cohort	<i>n</i>	Baseline level	Follow-up level	Change (SD)	Adjusted difference of changes**	<i>P</i> value (adjusted**)
HbA1c	Bisoprolol	4586	6.809	6.726	−0.083 (0.014)	−0.05 (−0.09, −0.01)	0.014*
	Reference	4586	6.857	6.819	−0.037 (0.013)		
FPG	Bisoprolol	3682	135.175	132.979	−2.196 (0.591)	−1.72 (−3.35, −0.1)	0.038*
	Reference	3682	133.769	133.14	−0.629 (0.565)		
Cholesterol	Bisoprolol	3769	183.71	171.502	−12.208 (0.687)	−9.12 (−10.94, −7.31)	<0.001*
	Reference	3769	180.97	177.44	−3.53 (0.601)		
LDL-Cholesterol	Bisoprolol	3207	105.562	95.059	−10.503 (0.616)	−7.89 (−9.53, −6.26)	<0.001*
	Reference	3207	103.7	100.783	−2.917 (0.544)		
HDL-Cholesterol	Bisoprolol	3713	46.668	45.559	−1.109 (0.11)	−0.94 (−1.24, −0.65)	<0.001*
	Reference	3713	47.301	47.125	−0.176 (0.101)		
Triglycerides	Bisoprolol	3765	163.838	161.988	−1.85 (1.438)	−2.72 (−6.7, 1.26)	0.18
	Reference	3765	157.545	157.858	0.313 (1.389)		

*Significantly different versus reference cohort 1

**Adjusted for CVD, other sinus arrhythmias, recent diagnosis of CVD and arrhythmia and accumulative BB dose

Table 2 Absolute change in measurements, bisoprolol vs. reference cohort 2 (carvedilol users)

Measurement	Cohort	<i>n</i>	Baseline level	Follow-up level	Change (SD)	Adjusted difference of changes**	<i>P</i> value (adjusted**)
HbA1c	Bisoprolol	154	7.04	6.987	−0.054 (0.07)	0.34 (0.04, 0.63)	0.024*
	Reference	154	7.266	6.876	−0.389 (0.13)		
FPG	Bisoprolol	123	141.099	145.465	4.366 (4.051)	6.95 (−4.19, 18.1)	0.22
	Reference	123	146.383	143.796	−2.586 (3.915)		
Cholesterol	Bisoprolol	134	173.195	164.991	−8.204 (3.228)	−0.46 (−10.43, 9.51)	0.927
	Reference	134	177.64	169.898	−7.742 (3.867)		
LDL-cholesterol	Bisoprolol	120	100.307	90.651	−9.656 (2.602)	−3.48 (−11.88, 4.91)	0.415
	Reference	120	104.298	98.125	−6.173 (3.344)		
HDL-cholesterol	Bisoprolol	133	44.244	43.278	−0.966 (0.55)	−0.41 (−1.98, 1.15)	0.603
	Reference	133	43.972	43.42	−0.552 (0.57)		
Triglycerides	Bisoprolol	133	154.527	162.054	7.527 (6.417)	5.46 (−16.43, 27.36)	0.624
	Reference	133	161.379	163.441	2.062 (8.996)		

*Significantly different versus reference cohort 2

**Adjusted for smoking status and recent diagnosis of HTN

bisoprolol users, baseline 6.809, follow-up 6.726, $p=0.014$). Furthermore, FPG was only significantly changed in reference cohort 1 compared to the bisoprolol cohort ($p=0.038$). The drop in total cholesterol was significantly greater in the bisoprolol cohort compared to reference cohorts 1 (no BB use, baseline 180.97, follow-up 177.44; bisoprolol use, baseline 183.71, follow-up 171.50, $p<0.001$) and 4 (nonselective BB users, baseline 182.83, follow-up 176.63; bisoprolol use, baseline 184.22, follow-up 172.16 $p=0.001$). Similarly, patients receiving bisoprolol had a significantly greater drop in LDL-C compared to cohort 1 (no BB use, baseline 103.70, follow-up 100.78; bisoprolol use, baseline 105.56, follow-up 95.06, $p<0.001$) and cohort 4 (nonselective BB use, baseline 104.90,

follow-up 100.81; bisoprolol use, baseline 106.36, follow-up 95.19, $p<0.001$). The drop in HDL-C with bisoprolol users compared to reference cohort 1 was significantly greater ($p<0.001$). There were no significant changes in triglyceride levels in any of the axes. Results of similar analysis with stratification for the first and last 6 months after the index can be found in Tables S3–S6 Supplementary materials.

Analysis of Bisoprolol on Glycemic Control and Identification of At-Risk Populations

The results of this secondary analysis indicated that none of the interactions between bisoprolol use and any

Table 3 Absolute change in measurements, bisoprolol vs. reference cohort 3 (β_1 selective BB users)

Measurement	Cohort	<i>n</i>	Baseline level	Follow-up level	Change (SD)	Adjusted difference of changes**	<i>P</i> value (adjusted**)
HbA1c	Bisoprolol	2142	6.946	6.866	−0.08 (0.021)	0.06 (−0.01,0.12)	0.078
	Reference	2142	7.031	6.894	−0.137 (0.025)		
FPG	Bisoprolol	1793	140.653	138.017	−2.636 (0.843)	1.9 (−0.52,4.33)	0.124
	Reference	1793	142.13	137.591	−4.539 (0.906)		
Cholesterol	Bisoprolol	1753	184.829	171.093	−13.736 (1.006)	−1.69 (−4.49,1.11)	0.237
	Reference	1753	189.667	177.62	−12.047 (1.013)		
LDL-cholesterol	Bisoprolol	1551	107.344	94.745	−12.599 (0.898)	−1.57 (−4.06,0.92)	0.216
	Reference	1551	112.635	101.606	−11.029 (0.895)		
HDL-cholesterol	Bisoprolol	1722	46.729	45.465	−1.264 (0.162)	−0.36 (−0.8,0.08)	0.113
	Reference	1722	46.633	45.728	−0.905 (0.158)		
Triglycerides	Bisoprolol	1746	168.812	163.788	−5.025 (2.199)	3.22 (−5.48,11.92)	0.468
	Reference	1746	173.867	165.624	−8.243 (3.875)		

*Significantly different versus reference cohort 3

**Adjusted for accumulative BB dose

Table 4 Absolute change in measurements, bisoprolol vs. reference cohort 4 (nonselective BB users)

Measurement	Cohort	<i>n</i>	Baseline level	Follow-up level	Change (SD)	Adjusted difference of changes**	<i>P</i> value (adjusted**)
HbA1c	Bisoprolol	1207	6.806	6.784	−0.022 (0.027)	0.07 (−0.0,0.15)	0.052
	Reference	1207	6.853	6.758	−0.095 (0.027)		
FPG	Bisoprolol	999	136.205	134.483	−1.722 (1.196)	0.54 (−2.74,3.82)	0.746
	Reference	999	138.056	135.791	−2.264 (1.168)		
Cholesterol	Bisoprolol	986	184.216	172.164	−12.052 (1.295)	−5.86 (−9.46, −2.25)	0.001*
	Reference	986	182.828	176.633	−6.195 (1.302)		
LDL-cholesterol	Bisoprolol	880	106.362	95.19	−11.172 (1.166)	−7.08 (−10.25, −3.92)	<0.001*
	Reference	880	104.904	100.814	−4.09 (1.114)		
HDL-cholesterol	Bisoprolol	971	46.758	45.621	−1.137 (0.203)	−0.43 (−1.03,0.18)	0.166
	Reference	971	48.03	47.32	−0.71 (0.232)		
Triglycerides	Bisoprolol	982	161.38	158.864	−2.516 (3.183)	2.19 (−5.62,10.01)	0.582
	Reference	982	152.947	148.237	−4.71 (2.396)		

*Significantly different versus reference cohort 4. **Adjusted for accumulative BB dose

Table 5 Bisoprolol users versus no BB users

	Coefficient [0.025 to 0.975]	Std err	<i>P</i> value
Intercept	2.7004 [2.595 to 2.806]	0.054	0
T2DM duration	0.0155 [0.012 to 0.019]	0.002	0
Sulfonylureas	0.3146 [0.268 to 0.361]	0.024	0
Baseline HbA1c	−0.4261 [−0.441 to −0.411]	0.008	0

covariates had a significant impact on the change in HbA1c. For bisoprolol versus no β -blocker users, no significant interaction effects were found with any covariates (Table 5). For bisoprolol versus carvedilol, interactions

between bisoprolol and obesity levels showed no significant impact on glycemic control, with coefficients for interaction terms not significant and *p* values greater than 0.1 (Table 6). Similarly, in comparisons of bisoprolol versus selective β -blockers (Table 7) and bisoprolol versus nonselective β -blockers (Table 8), no significant interaction effects between bisoprolol and covariates were observed, with all interaction terms showing nonsignificant *p* values. These findings reinforce that bisoprolol's effect on glycemic control is consistent across various demographic and clinical characteristics, with no evidence suggesting specific populations are at increased risk.

Table 6 Bisoprolol users versus carvedilol users

	Coefficient [CI 0.025 to 0.975]	std err	P value
Intercept	−0.8683 [−2.656 to 0.919]	0.9	0.337
Obesity = missing	−0.057 [−0.713 to 0.598]	0.33	0.862
Obesity = yes	0.32 [−0.132 to 0.784]	0.231	0.161
Bisoprolol	−0.13 [−2.268 to 1.991]	1.072	0.897
Age	0.013 [−0.012 to 0.038]	0.013	0.31
Smoking status	0.0103 [−0.337 to 0.357]	0.175	0.953
T2DM_duration	−0.0264 [−0.057 to 0.004]	0.015	0.086
HTN	0.1545 [−0.205 to 0.514]	0.181	0.396
CHF	0.2474 [−0.342 to 0.837]	0.297	0.407
Cancer	−0.5624 [−1.023 to −0.102]	0.232	0.017
Arrhythmias initiation	0.0731 [−0.263 to 0.409]	0.169	0.667
Metformin	−0.2719 [−0.604 to 0.06]	0.167	0.107
SGLT2	−0.1708 [−0.795 to 0.454]	0.314	0.588
Aspirin	−0.0192 [−0.353 to 0.315]	0.168	0.909
Dosage in defined daily dose	−0.0009 [−0.003 to 0.001]	0.001	0.403
Bisoprolol: age	0.011 [−0.018 to 0.04]	0.015	0.455
Bisoprolol: obesity = missing	−2.59E-16 [−2.9E-15 to 2.4E-15]	1.34E-15	0.847
Bisoprolol: obesity = yes	−0.582 [−1.17 to 0.006]	0.296	0.052

Table 7 Bisoprolol users versus other selective BB users

	Coefficient [CI 0.025 to 0.975]	Std err	P value
Intercept	3.1133 [2.954 to 3.272]	0.081	0
T2DM duration	0.0179 [0.012 to 0.024]	0.003	0
CHF	0.2734 [0.016 to 0.531]	0.131	0.038
CVD initiation	−0.1055 [−0.332 to 0.121]	0.115	0.36
CHF initiation	0.1518 [−0.114 to 0.418]	0.136	0.263
Sulfonylureas	0.2583 [0.194 to 0.323]	0.033	0
Baseline_HbA1c	−0.4879 [−0.51 to −0.465]	0.011	0

Table 8 Bisoprolol users versus other nonselective BB users

	Coefficient [CI 0.025 to 0.975]	Std err	P value
Intercept	2.8049 [2.594 to 3.015]	0.107	0
Obesity = no	−0.0946 [−0.183 to −0.006]	0.045	0.037
Obesity = yes	−0.1556 [−0.245 to −0.066]	0.046	0.001
T2DM duration	0.0203 [0.013 to 0.028]	0.004	0
CHF	0.8406 [0.555 to 1.126]	0.146	0
Arrhythmias initiation	−0.173 [−0.283 to −0.063]	0.056	0.002
DPP4-I	0.0866 [−0.006 to 0.179]	0.047	0.067
Baseline HbA1c	−0.4257 [−0.454 to −0.398]	0.014	0

Table 9 Proportion of hypoglycemic events

Study	Number of documented hypoglycemic events (%) bisoprolol cohort	Number of documented hypoglycemic events (%) reference cohort
1	30 (0.6%)	15 (0.3%)
2	< 10	< 10
3	19 (0.9%)	15 (0.7%)
4	12 (1.0%)	< 10

Proportion of Patients with Hypoglycemic Events

To determine the proportion of patients with hypoglycemic events, we counted the number of patients with at least one documented hypoglycemic event within each axis. The small number of documented hypoglycemic events across all comparisons precluded any meaningful statistical analysis. Results are reported in Table 9.

Discussion

Our results indicate that overall bisoprolol has a neutral to minimal effect on the glycemic profile across three out of the four groups. When comparing bisoprolol users to the BB-naïve population, a significant reduction in HbA1c and FPG among the bisoprolol users, though the delta was clinically negligible [13, 14]. Similarly, treatment with bisoprolol showed no clinically significant effect on lipid values,

though a statistically beneficial impact was demonstrated when comparing cholesterol deltas pre and post treatment to corresponding deltas of users of nonselective β -blockers [12, 13]. To our knowledge this is the first retrospective study using a detailed comparison of nonselective, and selective BBs to assess their impact on the glycemic and lipid profile of T2DM patients.

With the increasing prevalence of type 2 diabetes, the high risk of comorbid hypertension and high cardiovascular risk of diabetic patients, hypertensive patients with diabetes are candidates for immediate initiation of antihypertensive drug treatment together with lifestyle interventions. A large body of evidence supports the benefits of BP reduction in people with hypertension and type 2 diabetes to reduce macrovascular events and mortality, as well as to prevent microvascular complications, such as nephropathy and retinopathy. All major antihypertensive drug classes have shown to reduce CV outcomes in type 2 diabetes, including β -blockers [2]. However, the nonselectivity of first generation BB's has resulted in some reluctance to prescribe β -blockers in hypertensive diabetics, despite the guideline recommendations for their use. The detrimental effects of traditional BBs on glucose and lipid metabolism are thought to come about due to the lack of vasodilating properties, which creates a reduction in cardiac output, thereby decreasing glucose delivery to peripheral tissue. Yet, in a more recent study, bisoprolol was shown to increase insulin sensitivity and reduce fasting plasma glucose, as well as improve the TGs and cholesterol. This study was performed in a population with metabolic syndrome and hypertension and did not include patients with T2DM [15]. The purpose of our study was to run multiple comparisons between nonselective and selective β -blockers to affirm if this beneficial effect also applies to patients with T2DM.

Our results align with the existing literature, indicating that both carvedilol and bisoprolol have neutral effects on the glycemic profile and a favorable impact on the lipid profile, with bisoprolol being a highly selective β_1 blocker and carvedilol being a nonselective β and α blocker with additional vasodilatory properties [14–17]. The increased selectivity of β -blockers such as bisoprolol for β_1 -adrenergic receptor has improved therapeutic safety, especially for patients with T2DM [20].

Studies on bisoprolol showed a neutral effect on glucose and insulin levels after 24 weeks of treatment at 5 to 10 mg/day in 13 patients with hypertension [16]. Furthermore, in 1986, in patients with hypertension and T2DM, bisoprolol did not change blood glucose, HbA1c, and glucosuria when compared with placebo. No hypoglycemia was observed. Serum cholesterol and triglyceride levels also remained constant [19]. In patients with T2DM and systolic heart failure, as well as patients with mild to moderate HTN, bisoprolol use was not shown to worsen glycemic control

or negatively affect the lipid profile [17, 18]. A recent literature review of β -blockers in cardiac failure and associated blood glucose impairment, confirmed that bisoprolol, nebivolol, or carvedilol may be recommended as a drug of choice in patients with cardiac failure and impaired carbohydrate metabolism, due to their neutral effect on glycemic control, and their favorable effect on the lipid profile [21]. The ESH Guidelines support the use of β -blockers among the recommended first-line antihypertensive drugs, and are preferred drugs at any step in patients with heart failure with reduced left-ventricular ejection fraction, chronic coronary syndromes, or atrial fibrillation needing heart rate control [22]. Our secondary analysis confirms that bisoprolol does not have a differential impact on glycemic control in specific subpopulations defined by age, obesity, or baseline HbA1c levels. These findings reinforce that bisoprolol's effect on glycemic control is consistent across various demographic and clinical characteristics, with no evidence suggesting specific populations are at increased risk.

Our study has several limitations. First, this study was an observational retrospective study and not a randomized controlled trial, hence although matching was performed in an attempt to mitigate unobserved confounders, the choice of BB by the treating physician could be a source of bias. The axes which compared carvedilol users to bisoprolol users is the smallest one with only 156 pairs, far below the other axes (axes 1, $n=4672$, axes 3, $n=2168$, axes 4, 1224), as a result this axis may not have had sufficient power to detect statistical differences. In addition, this group was not balanced in terms of sex with a cohort that was 78.85% male. Furthermore, the distribution of participants to the different axes was based on drug purchase. This does not give us any information regarding medication adherence of the patients. Additionally, the follow-up period was 12 months, in a future study, it might be beneficial to follow these cohorts for a longer time period to check if the trends are stable. Finally, there may be confounding factors that affect glycemic control and metabolic lipid parameters such as dietary habits and physical activity that are not captured in our database and therefore not accounted for in our analysis.

Conclusions

This retrospective comparative study demonstrates that bisoprolol has a neutral effect on the glycemic profile. Additionally, bisoprolol showed a statistically beneficial impact on cholesterol when compared to no treatment and nonselective β -blockers, and is comparable to carvedilol. Bisoprolol's effect on glycemic control is consistent across various demographic and clinical characteristics, with no evidence suggesting specific populations are at increased risk. Bisoprolol, a highly selective β_1 blocker, should be considered in the

cardiovascular management of T2DM diabetic patients when indicated, without negatively affecting glycemic control or metabolic lipid parameters and have a potentially beneficial effect on cholesterol.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10557-025-07753-7>.

Author Contribution SG and TP conceptualized the study. YY was responsible for data curation. YY, MH, SG, and TP designed the study and methodology. YY was in charge of the formal analysis. SG was project administrator. TP provided supervision. YY did the visualization, and SG, YY, MH, and TP collaborated on validation. SG, YY, MH, and TP wrote the original draft, while all authors provided extensive comments during manuscript writing, review, and editing.

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Data Availability According to the Israel Ministry of Health regulations, individual level data cannot be shared openly. Specific requests for remote access to de-identified community level data should be referred to Kahn Sagol Maccabi Research and Innovation Centre, Maccabi Healthcare Services.

Code Availability Specific requests for remote access to the code used for data analysis should be referred to Kahn Sagol Maccabi Research and Innovation Centre, Maccabi Healthcare Services.

Declarations

Conflict of Interests The authors declare that they have no conflicts of interest in this work.

Ethics Approval This study was approved by the MHS Institutional Review Board (MHS-0001–24). Owing to the retrospective design of the study, informed consent was waived by the IRB and all identifying details of the participants were removed before computational analysis.

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